

GenAI: Requirement Analysis (in STLC)

This module connects **Generative AI** with the **Requirement Analysis** phase of **Software Testing Life Cycle (STLC)**.

Requirement Analysis in STLC

Requirement Analysis is the first phase of STLC where testers:

- Study business requirements
- Identify testable features
- Clarify ambiguities
- Define scope of testing

Objectives

- Understand what to test
 - Identify risks
 - Prepare test strategy
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Role of GenAI in Requirement Analysis: Clarifying Requirements

GenAI can:

- Rephrase vague requirements
- Identify missing details
- Ask clarifying questions
- Convert business language to technical language

Example

Input Requirement:

"System should be fast."

GenAI Clarification:

- What is acceptable response time?
- How many concurrent users?
- What operations must be fast?

Role of GenAI in Requirement Analysis: Suggesting Focus Areas

GenAI highlights:

- High-risk modules
- Security-sensitive areas
- Performance-critical features

Example

Requirement:

"Online payment feature"

GenAI suggests focus on:

- Payment gateway integration
- Data validation
- Error handling
- Security testing

Role of GenAI in Requirement Analysis: Generating Acceptance Criteria

Acceptance Criteria define **conditions to consider a feature complete**.

GenAI can:

- Generate Given-When-Then statements
 - Create edge-case criteria
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Generating Acceptance Criteria: Example

Feature: User Login

Input Prompt:

Generate acceptance criteria for login feature.

GenAI Output:

- Given valid credentials, when user logs in, then login is successful.
 - Given invalid password, when user logs in, then error message shown.
 - Given empty fields, when user logs in, then validation error shown.
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Demo: Analysing Requirements Using GitHub Copilot

Step 1: Create file

requirements.md

Step 2: Type Comment

```
<!-- Analyze requirements for online grocery app -->
```

Step 3: Copilot Suggests

```
## Functional Requirements
- User registration
- Login
- Search products
- Add to cart
- Checkout

## Non-Functional Requirements
- Performance
- Security
- Usability
```

Step 4: Ask for Acceptance Criteria

```
<!-- Generate acceptance criteria for checkout -->
```

Copilot generates scenarios.

Benefits of Using GenAI in Requirement Analysis

- Faster understanding
 - Better coverage
 - Reduced ambiguity
 - Improved collaboration
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Limitations

- May misinterpret domain logic
 - Requires human validation
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Quick Check

Q1. Which STLC phase involves understanding requirements?

Requirement Analysis

Q2. True/False

GenAI can generate acceptance criteria.

True

Q3. Best practice?

A) Accept AI output blindly

B) Review with stakeholders

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Summary

Generative AI enhances requirement analysis by:

- Clarifying vague requirements
- Suggesting focus areas
- Generating acceptance criteria